Underwater Acoustic Doppler Velocity Estimation Based on Matched Filtering

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***Abstract*—Utilizing the Doppler effect for velocity estimation is one of the key technologies in underwater acoustic sensing systems. High-precision velocity estimation is of great significance for applications such as underwater navigation, ocean current measurement, and sonar target tracking. However, in practical underwater environments, acoustic signals are often affected by factors such as ambient noise, propagation attenuation, and multipath effects, posing significant challenges to the accuracy and reliability of velocity estimation. Matched filtering, as a classical signal processing method, maximizes the output signal-to-noise ratio when the transmitted signal waveform is known. Therefore, this method has been widely applied in underwater acoustic detection and related parameter estimation fields.In this paper, a basic Doppler velocity estimation method based on matched filtering is investigated. First, the Doppler effect in underwater acoustic propagation is analyzed and a mathematical signal model is established. Then, a matched filtering receiver is designed to detect the echo signal and estimate the Doppler frequency shift. The velocity is calculated from the estimated Doppler shift. Simulation experiments are conducted using Linear Frequency Modulated (LFM) signals under different signal-to-noise ratio conditions. The results show that the proposed method can effectively estimate target velocity and maintain stable performance in moderate noise environments. The study demonstrates that matched filtering provides a reliable basis for underwater acoustic Doppler velocity estimation.**

***Index Terms*—underwater acoustics, Doppler effect, velocity estimation, matched filtering, Linear Frequency Modulated**

I. Introduction

Underwater acoustic velocity measurement is of great significance in the fields of underwater navigation, marine environment monitoring, target detection and tracking, and underwater acoustic communication systems. Electromagnetic waves propagate and attenuate severely in seawater. Therefore, sound waves are widely used for underwater information transmission and target detection due to their relatively low propagation loss. In many underwater applications, such as Doppler velocimetry, active sonar target tracking, and Autonomous Underwater Vehicle (AUV) navigation, accurate estimation of the relative velocity between the sound source and the target is required.

In recent years, scholars both domestically and internationally have conducted extensive research on the estimation of underwater acoustic Doppler. For example, Yang et al. proposed an improved Doppler sonar velocity estimation method based on accuracy evaluation and selection [1]. Nadav Cohen et al. proposed a Hybrid Neural Coupling (HNC) method for seamless Autonomous Underwater Vehicles (AUV) navigation with limited Doppler Velocity Log (DVL) measurements [2] [3]. And research on practical applications of speed measurement has also begun to combine Kalman filtering, frequency shift supervision, and other methods to improve the smoothness and stability of short-term speed measurement results [4] [5]. In underwater acoustic channels, multipath propagation and Doppler scale changes can seriously affect signal propagation characteristics, so accurate estimation of channel parameters is of great significance. Huang et al. proposed an iterative estimation method based on resampling matching, which constructs a Doppler scaling dictionary and sequentially extracts the dominant path to achieve multipath parameter estimation [6]. Sun et al. established a delay Doppler model using sparse representation and proposed the GSOMP algorithm to improve estimation accuracy and computational efficiency [7]. In addition, Van der Werf et al. studied parameter estimation methods based on LFM training sequences and compared the performance of fractional Fourier transform and matched filtering in delay Doppler estimation [8].

Overall, existing research has demonstrated the importance of underwater Doppler estimation, and has also shown that a single method often struggles to balance accuracy, complexity, and real-time performance in complex environments. In underwater acoustic communication systems, complex multipath propagation and Doppler effects can significantly affect signal transmission, making accurate channel parameter estimation particularly important. Shao et al. proposed a channel estimation method based on fractional Fourier transform (FrFT), which achieves joint estimation of multipath delay and Doppler scale by scanning the received signal with FrFT and extracting two-dimensional features using keypoint detection [9]. In addition, Li et al. proposed a Doppler estimation method based on Acoustic Frequency Comb (AFC), which uses frequency shift information of multiple frequency components for weighted statistics to achieve high-precision and low complexity Doppler factor estimation.Based on the above analysis, this article focuses on a matching filter based underwater Doppler velocity estimation method [10]. Considering the stability of matched filtering in known signal detection and the good pulse compression characteristics of LFM signals, this paper uses LFM signals as the transmission waveform to establish an underwater acoustic echo model with Doppler

frequency shift. The echo detection capability is enhanced by matched filtering, and Doppler frequency shift estimation is completed by combining frequency domain analysis [11] [12]. Based on this, the radial velocity calculation of the target is achieved. The focus of this article is not to propose complex new estimation algorithms, but to provide a set of structurally clear, physically interpretable, and applicable underwater Doppler velocimetry methods for basic research and engineering entry-level verification, starting from the basic model, processing flow, and simulation verification. This provides a reference for the improvement research of planar multipath, low signal-to-noise ratio, and time-varying channels in the future.

II. Underwater acoustic signal model

A. Transmitter signal model

In sonar speed measurement systems, the transmitted signal usually uses a modulated signal with good autocorrelation performance. For example, linear frequency modulated (LFM) signals. LFM signals are widely used in sonar systems, mainly due to their pulse compression characteristics, Doppler tolerance, and excellent anti-interference capabilities.

The general time-domain expression of LFM signal is

$$s(t) = A \cdot \mathrm{rect}\left(\frac{t}{T}\right) \cdot e^{j2\pi\left(f_0 t + \frac{1}{2}Kt^2\right)} \tag{1}$$

The instantaneous frequency of LFM signal is

$$f(t) = f_0 + kt \tag{2}$$

Therefore, LFM signals exhibit frequency modulation signals with linear frequency changes over time in the time domain. This enables the receiver to accurately determine the presence and location of a target by detecting the appearance of correlation peaks in strong noise backgrounds [13].

B. Acoustic propagation model

When a sound signal propagates to a moving target and reflects back to the receiving end, a propagation delay occurs. Assuming the target distance is in uppercase R, the propagation delay is

$$\tau = \frac{2R}{c} \tag{3}$$

In the ideal case without multipath propagation, the received signal model is:

$$r(t) = \alpha s(t-\tau) + n(t) \tag{4}$$

- α is propagation attenuation coefficient.
- $n(t)$ is ambient noise.

C. Doppler Effect Model

When there is relative motion between the target and the sonar system, the received signal will produce a Doppler frequency shift. For narrowband signals, the Doppler shift of the received signal can be expressed as:

$$f_d = \frac{2v}{c} f_0 \tag{5}$$

- v is target radial velocity.
- c is the speed of sound.
- f_0 is signal center frequency.

The received signal is represented as

$$r(t) = \alpha s(t-\tau) e^{j2\pi f_d t} + n(t) \tag{6}$$

D. Multipath propagation model

Sea surface reflection and seabed reflection are commonly present in marine environments, so the received signal usually contains multiple propagation paths. The multipath propagation model can be represented as

$$r(t) = \sum_{i=1}^{L} \alpha_i s\left(t-\tau_i\right) e^{j2\pi f_{d,i} t} + n(t) \tag{7}$$

Multipath interference is one of the main factors limiting the accuracy of Doppler estimation. In shallow or confined waters, multipath effects cause the echo from the sonar receiver to appear as a coherent superposition of multiple path components. This superposition will result in frequency selective fading, causing the spectral structure of the echo signal to be disrupted. For Doppler estimation, its accuracy highly depends on the precise extraction of the instantaneous frequency of the echo.

E. Matching filter processing model

In order to improve the detection capability of echo signals, matched filtering is usually used to process the received signal. Output model of matched filtering:

$$y(t) = \int r(\tau) s^*(\tau - t) d\tau \tag{8}$$

According to the theory of matched filtering, when the received signal contains echo components that match the waveform structure of the transmitted signal, the filter output will produce a significant correlation peak at the position corresponding to the delay of that component. Based on this principle, matched filtering can not only achieve optimal linear detection performance under additive noise background, that is, maximize the output signal-to-noise ratio to improve the detection probability of weak echo signals, but also provide high-precision initial estimation of the arrival time of the target echo at the time corresponding to its output peak.

III. Underwater Doppler velocity estimation algorithm based on matched filtering

A. Matching filter echo detection

The output of the received signal after passing through the matched filter is

$$y(t) = r(t) * h(t) = \int_{-\infty}^{+\infty} r(\lambda) s^*(\lambda - t) d\lambda \tag{9}$$

This equation is essentially equivalent to the cross-correlation operation between the received signal and the reference signal, so when the echo is aligned with the reference

waveform at a certain delay position, a significant peak will appear in the output.

After substituting the matching filter expression, we can obtain

$$y(t) = \alpha \int s(\lambda - \tau) e^{j2\pi f_d \lambda} s^*(\lambda - t) d\lambda + \tilde{n}(t) \tag{10}$$

Here, $\tilde{n}(t)$ represents the filtered noise term. It can be seen that the output peak is related not only to the signal time delay τ but also affected by the Doppler shift f_d. When $f_d = 0$, the output peak is at its maximum; when $f_d \neq 0$, a certain degree of mismatch and peak attenuation occurs. This is precisely the reason why further consideration of Doppler compensation or joint parameter estimation is necessary under broadband or high-speed motion conditions.

B. Doppler frequency shift estimation

In basic speed measurement systems, when the relative bandwidth of the signal is small and the target velocity is much smaller than the speed of sound, narrowband approximation can be used to consider the Doppler effect as frequency shift. At this point, the echo gating signal can be approximated as

$$z[n] \approx A[n] e^{j2\pi(f_0 + f_d)n/f_s} + v[n] \tag{11}$$

Estimate the main frequency position through Fourier transform.The N_{FFFT} point discrete Fourier transform is

$$Z[k] = \sum_{n=0}^{N_w - 1} z[n] e^{-j2\pi kn/N_{\text{FFT}}} \tag{12}$$

The corresponding frequency axis is

$$f_k = \frac{k}{N_{\text{FFT}}} f_s, \quad k = 0, 1, \ldots, N_{\text{FFT}} - 1 \tag{13}$$

The observation frequency corresponding to the spectral peak is

$$\hat{f}_r = f_{k_{\max}}, \quad k_{\max} = \arg\max_k |Z[k]| \tag{14}$$

The corresponding Doppler frequency shift estimation is

$$\hat{f}_d = \hat{f}_r - f_0 \tag{15}$$

Obtain velocity estimation from the two-way Doppler relationship of active sonar

$$\hat{v} = \frac{c f_d}{2 f_0} \tag{16}$$

The fundamental spectrum peak method remains one of the most common and easily reproducible speed measurement implementations.

C. Doppler search based on matched filter bank

Due to the Doppler frequency shift, there may be a mismatch between the reference signal and the echo. If a zero Doppler reference template is directly used, the peak value of the matched filter will decrease as f_d increases. To reduce the impact of mismatch, a set of reference signals with candidate frequency shift compensation can be constructed to form a matched filter bank.

Let the candidate Doppler set be

$$\mathcal{F} = \{f_{d,1}, f_{d,2}, \ldots, f_{d,M}\} \tag{17}$$

Then the mth compensation reference signal is written as

$$s_m(t) = s(t) e^{j2\pi f_{d,m} t} \tag{18}$$

The corresponding matched filter is

$$h_m(t) = s_m^*(-t) \tag{19}$$

The output of the mth matched filter is

$$y_m(t) = r(t) * h_m(t) \tag{20}$$

Define two-dimensional search statistics

$$\Lambda(\tau, f_{d,m}) = |y_m(\tau)| \tag{21}$$

The joint estimation of the final delay and Doppler is

$$\left(\hat{\tau}, \hat{f}_d\right) = \arg \max_{\tau, f_{d,m} \in \mathcal{F}} \Lambda(\tau, f_{d,m}) \tag{22}$$

The matched filtering speed measurement method has the advantages of clear structure, intuitive results, and easy implementation. For highly maneuverable targets, strong multipath shallow sea environments, or significant broadband scale distortion problems, further extension should be made to joint time delay Doppler estimation, scale compensation filter banks, or higher-order parameter tracking methods

IV. Simulation Results and Analysis

In order to verify the effectiveness of the proposed underwater Doppler velocity estimation method based on matched filtering, this paper uses MATLAB platform for simulation experiments.

A. Matched Filter Output

Firstly, analyze the detection capability of matched filtering on echo signals. Generate LFM signals for transmission in simulation, and construct received signals based on the established underwater acoustic propagation model, including propagation delay, Doppler frequency shift, and Gaussian white noise.The received signal is matched and filtered to obtain the output signal. Since matched filtering is equivalent to calculating the cross-correlation between the received signal and the reference signal, the output will produce a significant peak when the echo signal is aligned with the reference signal.The output result of matched filtering is shown in Fig. 1.

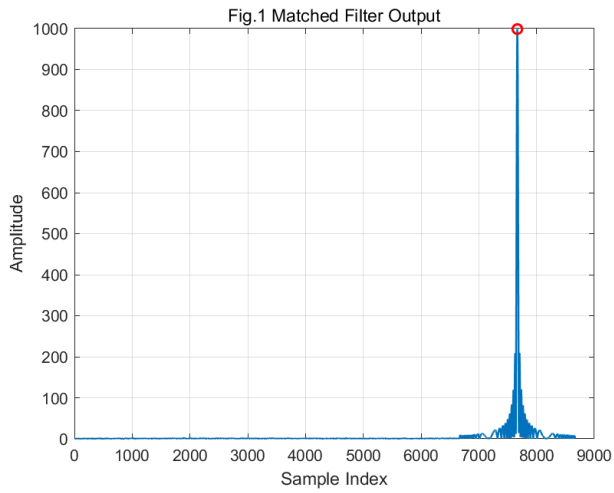

Fig. 1. Matched filter output

Figure 1 shows the output result of the matched filter, and it can be seen that there is a significant correlation peak at the target echo position, indicating that the transmitted signal and the echo signal are aligned at that moment. The sidelobes near the peak are typical sidelobe structures of LFM signal pulses after compression. This result indicates that matched filtering can effectively enhance the echo signal and improve the echo detection capability

B. Doppler Frequency Spectrum

After obtaining the main echo interval, FFT spectrum analysis is performed on the gated echo signal to estimate the Doppler frequency shift.

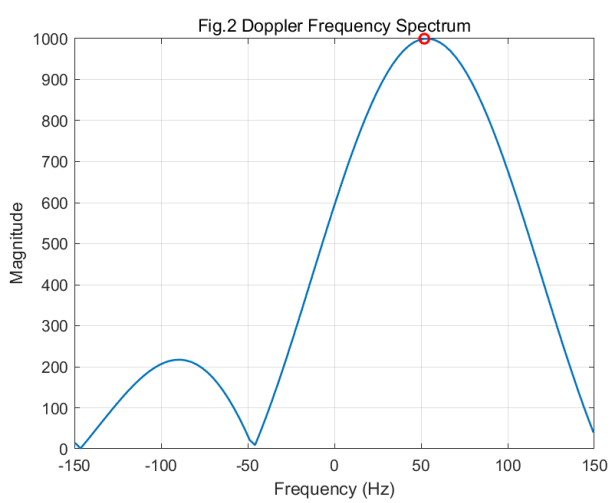

Fig. 2. Doppler frequency spectrum of received signal

Figure 2 shows the Doppler spectrum of the received signal. From the figure, it can be seen that there is a clear spectral peak around 53 Hz, which is consistent with the theoretical Doppler frequency shift. This indicates that spectral analysis can accurately extract target Doppler frequency information, thereby achieving target velocity estimation.

C. Velocity Estimation Error versus SNR

To further analyze the algorithm performance, speed measurement experiments were conducted under different signal-to-noise ratio conditions through simulation.And calculate the speed estimation error.The experimental results are shown in Fig. 3.

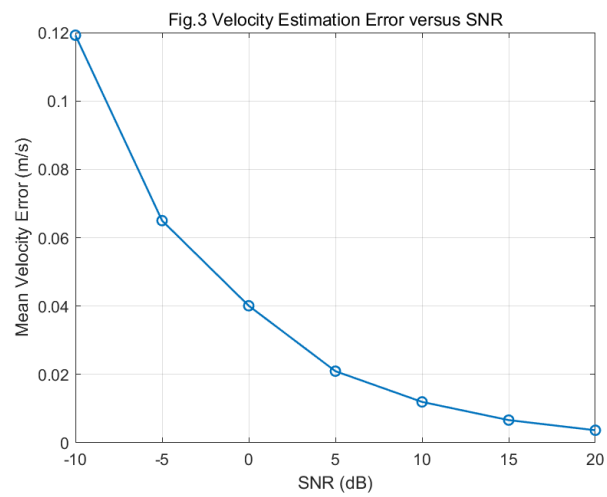

Fig. 3. Velocity estimation error versus SNR

From Figure 3, it can be observed that when the SNR is low, the velocity estimation error is relatively large due to the interference of noise on the spectral peaks; As the signal-to-noise ratio increases, the peak output of the matched filter becomes more prominent, the spectral peak detection becomes more stable, and the speed estimation error is significantly reduced.

D. Velocity Estimation for Different Target Speeds

To further validate the performance of the algorithm under different speed conditions, the simulation set the target speed range to 0-5 m/s and calculated the corresponding speed estimation results separately.

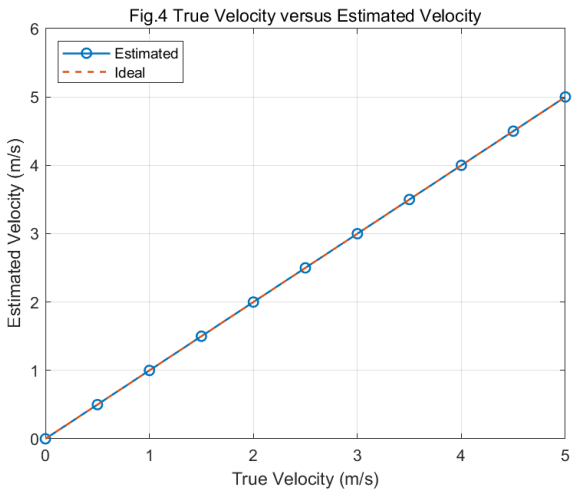

Fig. 4. True velocity versus estimated velocity

From Figure 4, it can be seen that there is a linear relationship between the estimated velocity and the true velocity, with most data points distributed near the ideal straight line. This indicates that the proposed method can achieve effective speed measurement over a wide speed range. At the same time, as the target velocity increases, the Doppler shift increases and the spectral peak becomes more pronounced, thus improving the stability of velocity estimation.

V. Conclusion

The following conclusions can be drawn from the above simulation results:Matched filtering can effectively improve the detection capability of echo signals and still produce significant correlation peaks under low signal-to-noise ratio conditions.By combining matched filtering detection with spectral analysis, stable estimation of Doppler frequency shift can be achieved. Under moderate signal-to-noise ratio conditions, the proposed speed measurement method can achieve smaller speed estimation errors.For low-speed moving targets, this method can still achieve stable speed measurement, indicating its suitability for typical underwater vehicle speed measurement scenarios.It should be pointed out that this simulation mainly considers single target and weak multipath environments. In complex marine environments, multipath propagation and time-varying channels may have an impact on velocity estimation performance. Therefore, future work will further investigate the Doppler velocity estimation problem under multipath conditions.

VI. Shortcomings and Improvements in Experiments

The simulation environment used in this article adopts an ideal underwater acoustic channel model, which only considers the influence of additive Gaussian white noise and does not take into account the complex propagation characteristics in real marine environments. In actual underwater acoustic channels, factors such as multipath propagation, sea surface reflection, seabed scattering, and changes in sound velocity profile can all affect the echo signal, leading to an increase in Doppler frequency shift estimation error [15]. Therefore, in future research, more realistic underwater acoustic channel models can be introduced [14].

In addition, this experiment only considers single target scenarios, and in practical underwater acoustic detection applications, there are often multiple moving targets. When multiple targets exist simultaneously, the Doppler frequencies of different targets may overlap with each other, thereby increasing the complexity of velocity estimation. Therefore, future research can further investigate Doppler separation and velocity estimation algorithms in multi-target environments to enhance their applicability in complex environments [16].

Finally, the target speed range in this simulation is 0-5 m/s, mainly targeting low-speed underwater moving targets. In practical applications, some underwater vehicles or mobile platforms may have higher motion speeds. Therefore, subsequent research can expand the simulation speed range and analyze the speed measurement performance under different speed conditions to further verify the applicability of the algorithm.

VII. Future Development Direction

Further introduction of more realistic underwater acoustic propagation channel models can enhance the practical reference value of experimental results. This article mainly uses an ideal underwater acoustic channel for simulation, and only considers the influence of additive Gaussian white noise. However, in real marine environments, the propagation of underwater acoustic signals is affected by various factors such as multipath propagation, sea and seabed reflections, changes in sound velocity profiles, and environmental noise, all of which can affect the structure of echo signals and Doppler estimation results. Therefore, in subsequent research, an underwater acoustic channel model that includes multipath propagation and environmental noise can be established to more comprehensively evaluate the performance of speed measurement algorithms in complex marine environments.

Modern signal processing and intelligent algorithms can be combined to further optimize speed measurement performance. In recent years, machine learning and deep learning methods have been widely applied in the field of signal processing. Future research can attempt to combine data-driven methods with traditional underwater acoustic signal processing techniques to achieve more robust Doppler frequency shift detection and target velocity estimation through intelligent algorithms, thereby further improving the overall performance of the system.

References

[1] Yang, Y.; Fang, S. Improved Velocity Estimation Method for Doppler Sonar Based on Accuracy Evaluation and Selection. J. Mar. Sci. Eng. 2021, 9 (6), 576.

[2] Cohen, N.; Klein, I. Seamless Underwater Navigation with Limited Doppler Velocity Log Measurements. IEEE Trans. Intell. Veh. 2024, 1–12.

[3] Cohen, N.; Klein, I. Enhancing Underwater Navigation through Cross-Correlation-Aware Deep INS/DVL Fusion. In OCEANS 2025 - Great Lakes; 2025; pp 1–6.
[4] Liu, P.; Kouguchi, N.; Li, Y. Velocity Measurement of Coherent Doppler Sonar Assisted by Frequency Shift, Kalman Filter and Linear Prediction. J. Mar. Sci. Eng. 2021, 9 (2), 109.
[5] Liu, P.; Liu, B.; Zhu, X.; Chen, P.; Li, Y. Noise Reduction of Velocity Measured by Frequency-Supervised Combined Doppler Sonar Using an Adaptive Sliding Window and Kalman Filter. J. Mar. Sci. Eng. 2024, 12 (12), 2320.
[6] Huang, S.; Fang, S.; Han, N. Iterative Matching-Based Parameter Estimation for Time-Scale Underwater Acoustic Multipath Echo. Appl. Acoust. 2020, 159, 107094.
[7] Sun, Q.; Wu, F.-Y.; Yang, K.; Ma, Y. Estimation of Multipath Delay-Doppler Parameters from Moving LFM Signals in Shallow Water. Ocean Eng. 2021, 232, 109125.
[8] Van Der Werf, I.; Hendriks, R. C.; Heusdens, R. Channel Parameter Estimation Using a Wideband LFM Preamble: Comparison of the Fractional Fourier Transform and Matched Filtering. In 2023 31st European Signal Processing Conference (EUSIPCO); IEEE: Helsinki, Finland, 2023; pp 1430–1434.
[9] Shao, J.; Zhang, C.; Xie, Y.; Mishra, D.; Rong, Y.; Chen, P.; Yuan, J. Fractional Fourier Transform Based Channel Estimation in Underwater Acoustic Communications. In OCEANS 2024 - Singapore; IEEE: Singapore, Singapore, 2024; pp 1–6.
[10] Qin, Z.; Li, C.; Wang, H.; Nie, L.; Yin, H. Active Underwater Acoustic Signal Detection Using Image Pattern Recognition Techniques. Ocean Eng. 2025, 341, 122417
[11] Han, Y.; Han, S.; Zhao, H.; Hu, Y.; Xu, J.; Yang, G. Doppler Compensation Techniques for M-Ary Sequence Spread Spectrum Signals Based on Correlation Cost Factors in Mobile Underwater Acoustic Communication. J. Mar. Sci. Eng. 2024, 12 (12), 2151.
[12] Abu, A.; Miskovic, N.; Chebotar, O.; Cukrov, N.; Diamant, R. Multiple Mobile Target Detection and Tracking in Active Sonar Array Using a Track-Before-Detect Approach. arXiv April 16, 2024.
[13] Lv, C.; Sun, Q.; Chen, H.; Xie, L. Doppler and Channel Estimation Using Superimposed Linear Frequency Modulation Preamble Signal for Underwater Acoustic Communication. J. Mar. Sci. Eng. 2024, 12 (2), 338.
[14] Li, J.; Qian, Z.; Hong, D.; Zhai, J. Precise and Low-Complexity Method for Underwater Doppler Estimation Based on Acoustic Frequency Comb Waveforms. Front. Mar. Sci. 2024, 11.
[15] Li, B.; Tong, F.; Li, J.; Zheng, S. Cross-Correlation Quasi-Gradient Doppler Estimation for Underwater Acoustic OFDM Mobile Communications. Appl. Acoust. 2022, 190, 108640.
[16] Song, C.; Jing, T. LRS: A Positive and Negative Linear Frequency Modulation Method for Ranging and Speed Measurement. In Review May 7, 2021.